

Theorem 1: The Riemann Relations as an Identity

symmetry and positivity of the period matrix

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1 Setting

The Riemann relations — symmetry and positive definiteness of the imaginary part of the period matrix — are proved on a curve *identically*, with no numerical check: a numerical control of an identity would be weaker than the proof. For the $N = 15/30$ stand this means: on the 91×182 and 406×812 matrices there is nothing to check numerically — it is proved.

Definition 1 (normalized period matrix). In the symplectic basis $\mathfrak{B} = \{a_1, \dots, a_g, b_1, \dots, b_g\}$ normalize the forms by $\int_{a_i} \omega_j = \delta_{ij}$ and set $\Omega_{ij} = \int_{b_i} \omega_j$.

Theorem 2 (Riemann relations). (1) $\Omega^T = \Omega$; (2) $\text{Im } \Omega$ is positive definite: for every $0 \neq \lambda \in \mathbb{R}^g$, $\lambda^T (\text{Im } \Omega) \lambda > 0$.

Proof. The tool is Riemann’s bilinear identity: for the fundamental $4g$ -gon F and closed forms α, β ,

$$\iint_F \alpha \wedge \beta = \sum_{i=1}^g \left(\int_{a_i} \alpha \int_{b_i} \beta - \int_{b_i} \alpha \int_{a_i} \beta \right).$$

Symmetry: at $\alpha = \omega_j, \beta = \omega_k$ the holomorphic $(2, 0)$ -object $\omega_j \wedge \omega_k$ vanishes ($\Omega_{C_N}^2 = 0$), giving $\Omega_{jk} - \Omega_{kj} = 0$. *Positivity:* at $\alpha = \omega, \beta = \bar{\omega}$ we have $\omega \wedge \bar{\omega} = -2i|g|^2 dx \wedge dy$; the left side is $-2i \int |g|^2 dx dy$; the right is $-2i \lambda^T (\text{Im } \Omega) \bar{\lambda}^T$ (symmetry of part 1 used). Canceling $-2i$: the quadratic form equals $\int_{C_N} |g|^2 dx dy > 0$. $\square$

Remark 3 (why no self-adjoint operator). “Symplectic” refers exclusively to the skew intersection form J on the integer lattice $H_1(C_N, \mathbb{Z})$ — a topological object. The theorem involves exactly two constructed objects: the cycle lattice and the integrals of forms. The Riemann relations are an identity that our own period matrix must satisfy; the numbers remain a one-off sanity line outside the text.

Summary of results: summary

The theorem is proved by Riemann’s bilinear identity; the only analytic premise is Stokes’ formula. Consequence: J is a polarization — the input of the Hodge lattice index module (theorem 3).

Verification: reproduction

The identity nature means: verification is a machine proof of structure, not a numerical run. Lean checks the integer corollaries (ranks 91, 406); the laboratory runs the sanity line (V7: rank = g , condition numbers 8.63, 18.17).